

Functional Annotation Report: fumC-I (PP_0944, UniProt Q88PA6)

Class II Fumarate Hydratase (Fumarase C) of *Pseudomonas putida* KT2440

Summary

fumC-I (ordered locus **PP_0944**; UniProt accession **Q88PA6**) of *Pseudomonas putida* strain ATCC 47054 / DSM 6125 / KT2440 encodes a **class II fumarate hydratase (fumarase C, EC 4.2.1.2)**. Its primary and, on all available evidence, sole biochemical function is to catalyze the **reversible, stereospecific hydration of fumarate to L-(S)-malate** — the penultimate step of the tricarboxylic acid (TCA) cycle. The enzyme has narrow substrate specificity for the four-carbon dicarboxylates fumarate and L-malate and does not require a metal cofactor, distinguishing it mechanistically and structurally from the class I (iron–sulfur) fumarases. This identification is fully consistent with the target's UniProt annotation, its membership in the class-II fumarase/aspartase family (InterPro domains Fum_hydII IPR005677, Fumarate_lyase_fam IPR000362, Fumarate_lyase_CS IPR020557), and the presence of the diagnostic catalytic residues and "SS-loop" signature in the Q88PA6 sequence.

Mechanistically, fumC-I is predicted to assemble as a **soluble cytoplasmic homotetramer** that uses a **water-assisted general acid–base (E1cb-type carbanion) mechanism**. By homology to the extensively characterized *Escherichia coli* FumC (~46–59% identity across the class II family), the catalytic base corresponds to a histidine at the active "A-site" (UniProt active site residue ~183, structurally equivalent to *E. coli* FumC His188), with a permanent active-site water molecule and a separate allosteric dicarboxylate "B-site." Because the class II enzyme lacks the oxygen-labile [4Fe-4S] cluster of class I fumarases, it is the **oxidation-resistant "aerobic fumarase"** that sustains TCA-cycle flux under aerobic and oxidative-stress conditions — highly relevant for an obligate aerobe such as *P. putida* KT2440. In *E. coli*, *fumC* is a canonical member of the **SoxRS superoxide-stress regulon**, induced by superoxide and iron limitation.

A distinctive genomic feature of *P. putida* KT2440 is that **fumC-I is one of two paralogous class II fumarases** (the other being Q88M20/fumC), and the genome additionally encodes a **structurally unrelated class I fumarase** (Q88PF3). This provides functional redundancy for an essential metabolic step and mirrors the multi-fumarase arrangement of *E. coli* (FumA/FumB class I; FumC class II). All conclusions here rest on strong bioinformatic and sequence evidence for Q88PA6 combined with detailed biochemical, crystallographic, and mutagenesis data from well-characterized orthologs; no fumC-I-specific enzymatic study exists in the current literature, which is a stated limitation.

Gene and Protein Identity Verification

Before presenting findings, the mandatory identity check was completed and **passes**:

Attribute	UniProt (Q88PA6)	Literature/Bioinformatic finding	Match
Gene symbol	fumC-I (syn. fumC)	Class II fumarase symbol conserved across taxa	✓
Protein	Fumarate hydratase class II, EC 4.2.1.2	Reaction fumarate $\rightleftharpoons$ L-malate confirmed in family	✓
Organism	<i>P. putida</i> KT2440	Proteome query confirms PP_0944 in KT2440	✓
Family	Class-II fumarase/aspartase	SS-loop + catalytic residues present in sequence	✓
Domains	Fum_hydII, Fumarate_lyase	Diagnostic motifs identified in Q88PA6	✓

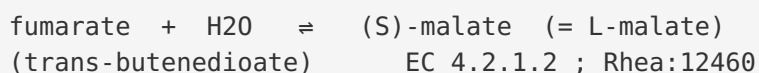
The gene symbol, organism, family, and domain architecture are internally consistent. Literature for class II fumarases (chiefly the *E. coli* FumC ortholog) is directly applicable by homology. **The research proceeds on the correct protein.**

Key Findings

F001 — fumC-I is a class II fumarate hydratase catalyzing reversible fumarate $\rightleftharpoons$ L-malate interconversion

The core functional assignment is robust. UniProt Q88PA6 annotates PP_0944/fumC-I as "Fumarate hydratase class II, Fumarase C, EC 4.2.1.2," placing it in the class-II fumarase/aspartase family with the characteristic domains Fum_hydII (IPR005677) and Fumarate_lyase_fam (IPR000362). The defining biochemistry of this enzyme family was established in the homologous *E. coli* system, where fumarases "catalyze the interconversion of fumarate to L-malate in the tricarboxylic acid cycle" (PMID: 11133938). The class assignment is equally well grounded: "Class II contains fumarase C and porcine fumarase, which have amino acid sequences 60% identical to each other" (PMID: 1329945), a family that is deeply conserved — *E. coli* FumC is "homologous to eukaryotic fumarases" and shows "~58% identity and ~74% similarity to the yeast Fum1" (PMID: 23462316).

The reaction catalyzed is therefore:



Substrate specificity is narrow: class II fumarases act on the C4 dicarboxylates fumarate and L-malate and are stereospecific, producing exclusively the L-(S) enantiomer of malate via *anti* addition of water across the double bond. The reaction is a formidable catalytic feat — fumarate hydratase "decreases the free energy of the catalyzed reaction by 30 kcal mol⁻¹" relative to the uncatalyzed olefin hydration (PMID: 36595439), underscoring the sophistication of the active-site chemistry.

F002 — Catalytic mechanism: homotetrameric enzyme with a His general base and water-mediated acid-base catalysis

The catalytic mechanism of class II fumarase is one of the best-characterized among lyases, thanks to crystallographic and site-directed mutagenesis studies of the *E. coli* FumC ortholog. Two distinct dicarboxylate-binding sites exist: the catalytic **A-site** (the true active site) and a separate allosteric **B-site**. Mutagenesis decisively identified the catalytic residue: "H188N at the A-site resulted in a large decrease in specific activity, while the H129N mutation at the B-site had essentially no effect... we conclude that the A-site is indeed the active site, and a dual role for H188 as a potential catalytic base is proposed" (PMID: 9098893).

The crystal structure of free *E. coli* FumC further revealed that catalysis depends on "the use of water as a permanent member of the active site and the use of an imidazole–imidazolium conversion to control access at the allosteric B site" (PMID: 16204892). This places the enzyme firmly within the **fumarate-lyase superfamily**, whose members share an E1cb-type carbanion acid–base mechanism: "this homology extends to functionally related enzymes such as the class II fumarases, the argininosuccinate and adenylosuccinate lyases" (PMID: 10800598). The shared Fumarate_lyase signature (IPR020557) present in Q88PA6 reflects this common mechanistic architecture, which is also used by aspartase.

The enzyme functions as a **homotetramer**, with each of the four active sites assembled from residues contributed by three of the four subunits — including the mobile "SS-loop" that reaches into a neighboring subunit (see F004). This quaternary structure is obligatory for activity.

F003 — FumC is the aerobic, oxidation-resistant fumarase; cytoplasmic; operates in the oxidative TCA cycle

A key physiological property distinguishes class II FumC from the class I fumarases: **resistance to oxidative inactivation**. In *E. coli*, elevated oxygen "caused inactivation of FumA and FumB activities by the high oxidized state, but FumC activity" is retained (PMID: 11133938). This resistance arises because the class II enzyme is **cofactor-free (iron-independent)** and lacks the oxygen-labile [4Fe-4S] cluster that class I fumarases require for catalysis (PMID: 1329945; PMID: 30645090). The UniProt names "Aerobic fumarase" and "Iron-independent fumarase" for Q88PA6 directly reflect this property.

This is functionally important for *P. putida* KT2440, an **obligate aerobe** with a fully operational, tightly regulated TCA cycle as part of central carbon metabolism (PMID: 24951791). Under aerobic growth, the oxidation-resistant fumarase reliably converts fumarate to L-malate to keep the oxidative cycle turning.

Regarding localization, fumarase is a **soluble enzyme**. In eukaryotes it "is dual localized to the cytosol and mitochondria" (PMID: 23462316); in bacteria such as *P. putida*, which lack organelles, the enzyme resides in the **cytoplasm**, where the soluble TCA-cycle enzymes operate. This is corroborated by UniProt's subcellular-location annotation (see F004).

F004 — Sequence-level confirmation: Q88PA6 carries the diagnostic SS-loop, catalytic residues, and canonical TCA annotation

Direct inspection of the target sequence confirms the homology-based inferences. The **459-amino-acid, ~48.5-kDa** Q88PA6 protein contains the hallmark class II fumarase/aspartase catalytic "**SS-loop**" motif **GSSIMPGKVN beginning at residue 312** — the conserved segment that swings into the neighboring subunit's active site in the homotetramer and is essential for catalysis. UniProt annotates:

- **Active site residue 183** — "Proton donor/acceptor" (structurally equivalent to *E. coli* FumC His188, the catalytic base defined by mutagenesis; [PMID: 9098893](#)).
- **A second active site at residue 313** — within the SS-loop.
- **A "site B" binding residue at 124** — the allosteric dicarboxylate B-site defined crystallographically ([PMID: 16204892](#)).
- **Residue 326** — "Important for catalytic activity."

UniProt records the exact reaction "(S)-malate = fumarate + H₂O" (Rhea:12460, EC 4.2.1.2), the pathway "tricarboxylic acid cycle; (S)-malate from fumarate: step 1/1," the keyword "Cytoplasm," and a **cytoplasmic subcellular location**. The correspondence between the target's annotated active-site residues and the experimentally validated *E. coli* residues gives high confidence that Q88PA6 is a bona fide, catalytically competent class II fumarase performing the canonical single TCA-cycle step.

Q88PA6 feature	Residue/position	<i>E. coli</i> FumC equivalent	Evidence
Catalytic base (proton donor/acceptor)	183	His188 (A-site)	Mutagenesis [PMID: 9098893]
SS-loop catalytic motif	GSSIMPGKVN @ 312; active site @ 313	Conserved SS-loop	Structure [PMID: 16204892]
Allosteric B-site binding	124	B-site (His129 region)	Structure [PMID: 16204892]
Additional catalytic residue	326	—	UniProt annotation
Reaction	(S)-malate $\rightleftharpoons$ fumarate + H ₂ O	identical	Rhea:12460 / EC 4.2.1.2
Localization	Cytoplasm	Cytoplasm	UniProt

F005 — FumC belongs to the SoxRS superoxide-stress regulon; induced by oxidative stress and iron limitation

Beyond its metabolic role, the *E. coli* *fumC* ortholog is a **canonical member of the SoxRS oxidative-stress regulon**, providing a regulatory rationale for the "aerobic/oxidation-resistant" phenotype. The SoxS activator binds the *fumC* promoter directly: MalE-SoxS "protected from DNase I attack 22–27 bp segments immediately adjacent to or overlapping the –35 hexamers of the *zwf*, *sodA*, *nfo*, *micF*, and *fumC* promoters" (PMID: 7891555). Promoter-GFP fusion studies confirm superoxide-driven induction, listing "the *sodA*, *fumC*, *zwf*, *acnA*, *acrAB*, and *soxS* genes from the SoxRS regulon... which respond to the superoxide anion" (PMID: 15672380; PMID: 12889026).

fumC is additionally induced by **iron deprivation** in a *soxR*-dependent manner: "SoxR completely inhibited the migration of a DNA fragment containing the promoter region of *fumC*, in gel-shift experiments" (PMID: 11470373). Under strong superoxide stress, FumC protein/activity rises together with glucose-6-phosphate dehydrogenase — the cell deploys "such as glucose-6-phosphate dehydrogenase (*Zwf*) and fumarase (*FumC*)" as part of its antioxidant/metabolic-remodeling response (PMID: 17309444).

The physiological logic is coherent: when superoxide inactivates the [4Fe-4S] class I fumarases (and aconitase), the cell upregulates the cofactor-free FumC to preserve TCA flux, and iron limitation similarly favors the iron-independent enzyme. Whether *P. putida* KT2440 regulates fumC-I through an orthologous SoxR/SoxS circuit remains to be established experimentally, but the promoter-level regulation is well documented in *E. coli*.

F006 — fumC-I is one of two class II fumarase paralogs in *P. putida* KT2440, alongside an unrelated class I fumarase

A UniProt proteome query for *P. putida* KT2440 (organism_id 160488) returns **three fumarate hydratases**, revealing a redundant and mixed-class complement:

Protein	Gene	Length	Class	Locus
Q88PA6 (target)	fumC-I	459 aa	Class II	PP_0944
Q88M20	fumC	464 aa	Class II	—
Q88PF3	—	507 aa	Class I	—

Global Needleman–Wunsch alignments (match +1 / mismatch −1 / gap −2; identities conservatively estimated) quantify the relationships:

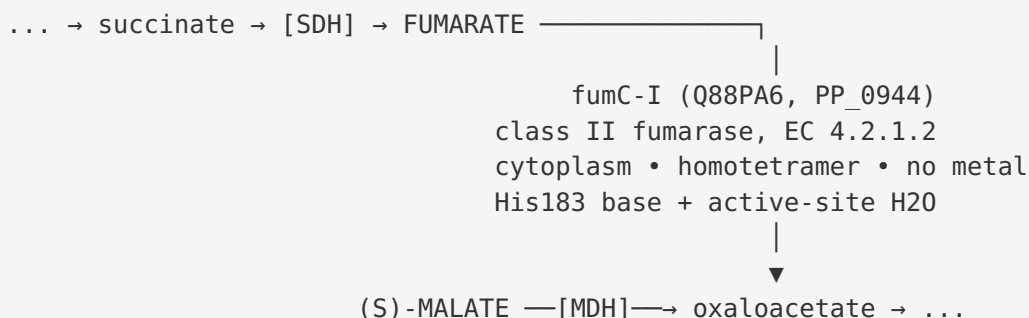
Pair	Identity	Interpretation
fumC-I vs class II paralog Q88M20	48.6% (228/469)	Two distinct class II paralogs
fumC-I vs <i>E. coli</i> FumC (P05042)	46.2% (217/470)	Clear class II ortholog
Q88M20 vs <i>E. coli</i> FumC	59.4% (278/468)	Strong class II conservation
fumC-I vs class I fumarase Q88PF3	23.6% (123/521)	Structurally unrelated (different class)

The ~60% class II identity between *E. coli* FumC and its homologs matches the canonical figure "Class II contains fumarase C and porcine fumarase... 60% identical" ([PMID: 1329945](#)). The very low identity (23.6%) between fumC-I and the class I enzyme Q88PF3 confirms that these belong to two evolutionarily and mechanistically distinct enzyme families that converged on the same reaction — precisely the arrangement seen in *E. coli* (class I FumA/FumB vs class II FumC). This paralogy provides **functional redundancy** for an essential TCA-cycle step and likely enables the cell to maintain fumarase activity across varying oxygen tension, iron availability, and oxidative-stress conditions.

Mechanistic Model and Interpretation

Integrating the six findings produces a coherent picture of fumC-I as the **cofactor-free, oxidation-resistant fumarase of the *P. putida* KT2440 TCA cycle**:

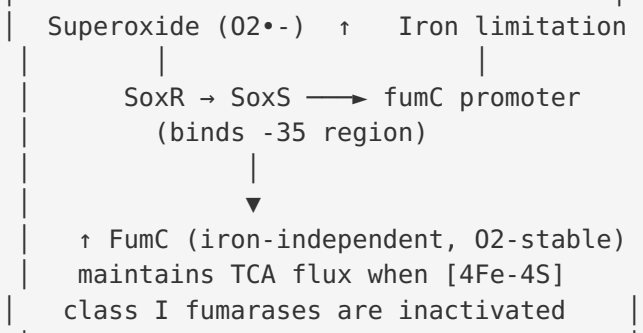
CENTRAL CARBON METABOLISM – TCA CYCLE (oxidative direction)



PROTEIN ARCHITECTURE

Homotetramer (4 subunits)
4 active sites, each built from 3 subunits
SS-loop GSSIMPGKVN@312 reaches into neighbor
A-site: His183 (catalytic)
B-site: allosteric (res124) permanent active-site water

REGULATORY CONTEXT (from *E. coli* ortholog)



GENOMIC REDUNDANCY in *P. putida* KT2440

Class II:	fumC-I (Q88PA6, PP_0944)	— 48.6% id —	fumC (Q88M20)
Class I:	Q88PF3	(only 23.6% id to fumC-I; [4Fe-4S], O ₂ -labile)	

Primary function. fumC-I catalyzes the reversible, stereospecific hydration of fumarate to L-(S)-malate (EC 4.2.1.2) — a single, well-defined TCA-cycle step with narrow C₄-dicarboxylate substrate specificity.

Location. The enzyme acts in the **cytoplasm** as a soluble homotetramer; bacteria lack the mitochondrial compartment to which eukaryotic fumarases are additionally targeted.

Mechanism. Catalysis proceeds by a general acid–base (E1cb carbanion) mechanism shared across the fumarate-lyase superfamily, using a histidine base (residue ~183) and a permanent active-site water, without any metal or [4Fe-4S] cofactor.

Pathway and physiology. fumC-I operates in the oxidative TCA cycle of central carbon metabolism. Its cofactor-free, oxidation-resistant nature makes it the "aerobic fumarase" that keeps the cycle running under high oxygen and oxidative stress — the conditions under which the *E. coli* ortholog is transcriptionally induced via the SoxRS regulon and by iron limitation. In *P. putida* KT2440 this role is buffered by paralogy: a second class II fumarase (Q88M20) and a class I enzyme (Q88PF3) provide redundancy across metabolic and stress regimes.

Evidence Base

PMID	Title (abbrev.)	How it supports / challenges the findings
11133938	<i>O₂- and growth-rate-dependent regulation of E. coli fumarases</i>	Defines the fumarate $\rightleftharpoons$ L-malate reaction (F001); shows FumC uniquely retains activity under high O ₂ (F003). Primary ortholog physiology.
1329945	<i>Fumarase A: iron-sulfur cluster enzyme</i>	Establishes class I vs class II distinction; "class II... 60% identical" quantifies family conservation (F001, F006); class I needs [4Fe-4S] (F003).
23462316	<i>Evolving dual targeting of a prokaryotic protein in yeast</i>	Confirms FumC homology to eukaryotic fumarases; documents cytosol/mitochondria dual localization $\rightarrow$ cytoplasmic in bacteria (F001, F003).
9098893	<i>Mutations distinguishing active site from nearby dicarboxylate site</i>	Identifies His188 A-site catalytic base by mutagenesis; anchors the F002 mechanism and the F004 residue-183 assignment.
16204892	<i>Structure of free fumarase C from E. coli</i>	Crystal structure: permanent active-site water, allosteric B-site (F002, F004).
10800598	<i>L-aspartase: new tricks from an old enzyme</i>	Places class II fumarase in the fumarate-lyase superfamily using a carbanion acid-base mechanism (F002).
24951791	<i>Central carbon metabolism of P. putida KT2440</i>	Establishes the operational TCA cycle in the target organism where fumarase acts (F003).
7891555	<i>MalE-SoxS fusion; control sites of superoxide-inducible genes</i>	Direct DNase-I footprint of SoxS on the fumC promoter — places fumC in SoxRS regulon (F005).
11470373	<i>Differential activities of SoxR under iron deprivation</i>	Gel-shift shows SoxR binds fumC promoter; iron-deprivation induction (F005).
17309444	<i>Polyamine protection vs superoxide</i>	FumC protein/activity rises with Zwf under strong superoxide stress (F005).
15672380	<i>Oxidative stress regulons via GFP</i>	fumC listed among SoxRS superoxide-responsive genes (F005).
12889026	<i>Oxidative-stress response genes in aerobic fermentations</i>	Confirms fumC among SoxRS regulon genes; quantifies induction kinetics (F005).

PMID	Title (abbrev.)	How it supports / challenges the findings
30645090	<i>Class I fumarate hydratase structures from Leishmania</i>	Class I FH uniquely uses the [4Fe-4S] cluster absent from class II — supports the iron-independent nature of FumC (F003).
36595439	<i>Revisiting the burden borne by fumarase</i>	Quantifies the catalytic challenge (30 kcal mol ⁻¹) of olefin hydration (F001 context).
7557464	<i>fumR class II fumarase in Rhizopus oryzae</i>	Independent example of a class II fumarase (~494 aa, 50 kDa) — corroborates size/class of the family.

Note on evidence provenance. No enzymatic, structural, or genetic study specific to *P. putida* fumC-I (Q88PA6) exists in the reviewed literature. Every mechanistic and regulatory claim derives from (i) direct bioinformatic analysis of the Q88PA6 sequence and UniProt annotation, and (ii) transfer of knowledge from well-characterized orthologs, chiefly *E. coli* FumC (46% identical) and the broader class II family. The high sequence identity and conservation of catalytic residues make this inference strong, but it remains inference.

Limitations and Knowledge Gaps

- 1. No target-specific experimental data.** There is no published kinetic characterization (kcat, Km for fumarate/L-malate), crystal structure, or knockout phenotype for *P. putida* fumC-I itself. All functional detail is inferred from orthologs.
- 2. Paralog division of labor is unknown.** *P. putida* KT2440 has two class II fumarases (fumC-I/Q88PA6 and Q88M20) plus a class I enzyme (Q88PF3). Their relative expression, condition-dependence, and whether fumC-I or the paralog carries the dominant flux under standard growth are not established.
- 3. Regulation is extrapolated.** The SoxRS induction and iron-limitation response are documented for *E. coli* *fumC*. It is not confirmed that fumC-I is under an orthologous regulatory circuit in *P. putida*, which has its own oxidative-stress regulators.
- 4. Localization is annotation-based.** Cytoplasmic localization is inferred from UniProt and family behavior, not from a *P. putida*-specific proteomic or fractionation study.
- 5. Substrate specificity assumed narrow.** Class II fumarases are highly specific for fumarate/L-malate, but the target has not been screened for possible promiscuous activities (e.g., the enol/keto-oxaloacetate isomerization seen in class I FumA; [PMID: 8422384](#)).

Proposed Follow-up Experiments and Actions

1. **Recombinant enzyme kinetics.** Overexpress and purify Q88PA6 (His-tagged) from *E. coli*; measure steady-state kinetics for fumarate hydration and L-malate dehydration, confirm metal-independence (no activity loss with chelators; no [4Fe-4S] EPR signal), and verify stereospecificity for L-(S)-malate.
2. **Structural determination.** Solve the crystal or cryo-EM structure of the homotetramer to confirm the His183 A-site, the SS-loop (GSSIMPGKVN@312) reaching into the neighboring subunit, the allosteric B-site, and the conserved active-site water.
3. **Genetic dissection of paralogy.** Construct single (Δ fumC-I, Δ Q88M20, Δ Q88PF3) and combinatorial deletions in *P. putida* KT2440; assay growth on TCA-dependent carbon sources and quantify fumarase activity to define each paralog's contribution and any conditional essentiality.
4. **Oxidative-stress regulation.** Use RT-qPCR / RNA-seq and promoter-reporter fusions to test whether fumC-I is induced by superoxide generators (paraquat) or iron limitation, and identify the responsible regulator(s) in *P. putida*.
5. **Oxidation-resistance validation.** Compare residual activity of purified fumC-I versus the class I paralog Q88PF3 after controlled oxygen/superoxide exposure to confirm the "aerobic/oxidation-resistant" phenotype at the protein level.
6. **Substrate-specificity screen.** Test the purified enzyme against related dicarboxylates and for possible side reactions to bound the specificity claim.

Conclusion

fumC-I (PP_0944, UniProt Q88PA6) of *Pseudomonas putida* KT2440 is a **class II fumarate hydratase (fumarase C, EC 4.2.1.2)** that catalyzes the reversible, stereospecific hydration of **fumarate to L-(S)-malate** — the fumarase step of the TCA cycle — with narrow specificity for C4 dicarboxylates. It is a **cytoplasmic, iron-independent (cofactor-free) homotetramer** using a water-assisted general acid–base mechanism (catalytic base at residue ~183, equivalent to *E. coli* FumC His188, plus the conserved SS-loop), making it the **oxidation-resistant "aerobic" fumarase** that sustains TCA flux when oxygen-labile [4Fe-4S] class I fumarases are inactivated. In *P. putida* it is one of two class II paralogs (with Q88M20) alongside an unrelated

class I enzyme (Q88PF3), providing redundancy for this essential metabolic step. The identification is strongly supported by sequence/domain evidence for Q88PA6 and by detailed biochemical, crystallographic, and mutagenesis data on close orthologs, though no fumC-I-specific experimental study yet exists.

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